Among them, the definition by Croxton and Cowden is considered as the most preferable one due to its comprehensiveness. It is clear from this definition that statistics brings out the following characteristics.

Characteristics of Statistics:

(1) Aggregate of facts collected in systematic manner for a specific purpose.

Statistics deals with the aggregate of facts and figures. A single number cannot be called as statistics. For example, the weight of a person with 65kg is not statistics but the weights of a class of 60 persons is statistics, since they can be studied together and meaningful comparisons are made in relation to the other. This reminds us of Joseph Stalin's well known quote, "One death is a tragedy; a million is a statistics." Further the purpose for which the data is collected is to be made clear, otherwise the whole exercise will be futile. The data so collected must be in a systematic way and should not be haphazard.

(2) Affected by large number of causes to marked extent.

Statistical data so collected should be affected by various factors at the same time. This will help the statistician to identify the factors that influence the statistics. For example, the sales of commodities in the market are affected by causes such as supply, demand, and import quality etc. Similarly, as mentioned earlier if a million deaths occur the policy makers will be immediately in action to find out the causes for these deaths to see that such events will not occur.

(3) Numerically expressed.

The statistical facts and figures are collected numerically for meaningful inference. For instance, the service provided by a telephone company may be classified as poor, average, good, very good and excellent. They are qualitative in nature and cannot be called statistics. They should be expressed numerically such as 0 to denote poor,1 for average, 2 for good, 4 to denote very good and 5 for excellent. Then this can be regarded as statistics and is suitable for analysis. The other types of quality characteristics such as honesty, beauty, intelligence, defective etc which cannot be measured numerically cannot be called statistics. They should be suitably expressed in the form of numbers so that they are called statistics.

(4) Enumerated or estimated with a reasonable degree of accuracy.

The numerical data are collected by counting, measuring or by estimating. For example, to find out the number of patients admitted in a hospital, data is collected by actual counting or to find out the obesity of patients, data are collected by actual measurements

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